**Step 5 : Calculation of test statistic**

To find sample mean and sample standard deviation:

| $x_i$ | $u_i = x_i - \bar{x}$ $(\bar{x} = 7)$ | $u_i^2$ | $y_i$ | $v_i = y_i - \bar{y}$ $(\bar{y} = 5)$ | $v_i^2$ |
|---|---|---|---|---|---|
| 6 | -1 | 1 | 5 | 0 | 0 |
| 7 | 0 | 0 | 6 | 1 | 1 |
| 9 | 2 | 4 | 5 | 0 | 0 |
| 2 | -5 | 25 | 7 | 2 | 4 |
| 13 | 6 | 36 | 1 | -4 | 16 |
| 3 | -4 | 16 | 7 | 2 | 4 |
| 4 | -3 | 9 | 2 | -3 | 9 |
| 8 | 1 | 1 | 7 | 2 | 4 |
| 7 | 0 | 0 | | | |
| 11 | 4 | 16 | | | |
| $\sum_{i=1}^{10} x_i = 70$ | $\sum_{i=1}^{10} u_i = 0$ | $\sum_{i=1}^{10} u_i^2 = 108$ | $\sum_{i=1}^{8} y_i = 40$ | $\sum_{i=1}^{8} v_i = 0$ | $\sum_{i=1}^{8} v_i^2 = 38$ |

**To find sample means:**

Let $(x_1, x_2, ..., x_{10})$ and $(y_1, y_2, ..., y_8)$ denote the scores of students in Batch I and Batch II respectively.

$$\bar{x} = \frac{\sum_{i=1}^{10} x_i}{10} = \frac{70}{10} = 7$$

$$\bar{y} = \frac{\sum_{i=1}^{8} y_i}{8} = \frac{40}{8} = 5$$

**To find combined sample standard deviation:**

$$s_X^2 = \frac{1}{9}\sum_{i=1}^{10}(x_i - \bar{x})^2 = \frac{1}{9}\sum_{i=1}^{10} u_i^2 = \frac{108}{9} = 12$$

$$s_Y^2 = \frac{1}{7}\sum_{i=1}^{8}(y_i - \bar{y})^2 = \frac{1}{7}\sum_{i=1}^{8} v_i^2 = \frac{38}{7} = 5.4$$

Pooled standard deviation is:

$$S_p = \sqrt{\frac{(m-1)s_x^2 + (n-1)s_y^2}{m+n-2}} = \sqrt{\frac{108+38}{10+8-2}} = \sqrt{9.125} = 3.021$$

The value of $T$ is calculated for the given information as

$$t_0 = \frac{\bar{x} - \bar{y}}{s_p\sqrt{\frac{1}{m} + \frac{1}{n}}} = \frac{7-5}{3.021\sqrt{\frac{1}{10} + \frac{1}{8}}} = 1.3957$$



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